1. Improving Care and Promoting Health in Populations: Standards of Care in Diabetes—2026

*American Diabetes Association Professional Practice Committee for Diabetes**

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The American Diabetes Association (ADA) "Standards of Care in Diabetes" includes the ADA's current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA's clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

SYSTEMS TO SUPPORT DIABETES POPULATION HEALTH

Recommendations

1.1 Ensure treatment decisions are timely, rely on evidence-based guidelines, address social determinants of health, and incorporate shared decision-making based on individual values, preferences, prognoses, comorbidities, and informed financial considerations. **B**

1.2 Align approaches to diabetes management with evidence-based care models. These models emphasize person-centered team care, integrated long-term treatment approaches to diabetes and comorbidities, and ongoing collaborative communication and goal setting among all team members and with people with diabetes. **A**

1.3 Care systems should facilitate in-person and virtual team-based care, include those knowledgeable and experienced in diabetes management as part of the team, and utilize patient registries, decision support tools, proactive care planning, and community involvement to meet needs of individuals with diabetes. **B**

1.4 Assess diabetes management, risk factors, and complications using reliable and relevant data metrics to improve processes of care and health outcomes, with attention to individual values, preferences, goals for care, and treatment burden, including costs of care. **B**

1.5 Health systems should adopt a culture of continuous quality improvement, implement benchmarking programs, and engage interprofessional teams to support sustainable and scalable process changes to improve quality of care and health outcomes. **A**

**A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at https://doi.org/10.2337/dc26-SINT.*

Duality of interest information for each contributor is available at https://doi.org/10.2337/dc26-SDIS.

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Diabetes and Population Health

Population health is a multidimensional concept that refers to the health outcomes of a group of individuals and the potentially varied distribution of those outcomes within the group (1). These outcomes can be measured in terms of health indicators (mortality, morbidity, quality of life, and functional status), disease epidemiology (incidence and prevalence), and behavioral and metabolic factors (physical activity, nutrition, A1C, time in range, etc.) (1). In clinical practice, population health commonly refers to the measurement and improvement of health outcomes for a defined group of individuals, often those receiving care within a particular health system or practice, and is central to operational "population health management," or the proactive, team-based, and data-driven approaches to improving health care experiences and team well-being while minimizing costs and addressing inequities, in alignment with the Quadruple Aim (2). More broadly, population health refers to the health of all people living in a geographic area or community, and it centers on upstream drivers of health that must be addressed, often through cross-sector collaboration and policy, to improve the broader conditions that influence health.

Clinical practice recommendations are tools for health care professionals who seek to improve health across populations; however, for optimal outcomes, diabetes care must also be individualized for each person with diabetes and for each person's context, as well as across the life span. Thus, efforts to improve population health will require a combination of policy-level, system-level, and person-level approaches. With such an integrated approach in mind, the American Diabetes Association (ADA) highlights the importance of person-centered care, defined as care that considers an individual's comorbidities and prognoses; is respectful of and responsive to individual preferences, needs, and values; and ensures that the individual's values guide all clinical decisions (3). Social determinants of health (SDOH)—factors often beyond an individual's direct control and potentially representing lifelong risks—play a significant role in both clinical and psychosocial outcomes. SDOH are defined as the economic, environmental, political, and social conditions in which people live, and their uneven distribution is responsible for a major part of health inequality worldwide (4). To improve health, support overall well-being, and eliminate disparities, it is crucial to address these determinants, particularly for individuals from racial and ethnic minoritized communities, underserved geographic areas (rural or urban), and those facing socioeconomic barriers to care and health (5). This section discusses the current state of diabetes and diabetes care in the U.S. and provides guidance for health care professionals as well as health systems, community partners, payors, and policymakers on improving the delivery of diabetes care to improve the health of all people at risk for or living with diabetes.

To provide actionable guidance for improving the health outcomes of people with or at risk for diabetes, this section examines care delivery and payment models demonstrated to support high-quality, evidence-based care; offers guidance on practical strategies for system-level improvement, including identifying and addressing gaps in diabetes care and outcomes experienced by population subgroups; and discusses opportunities to expand access to health care and diabetes self-management education and support (DSMES) through telehealth, mobile platforms, interprofessional team care, and engagement of community-based care partners and resources.

State of Diabetes and Diabetes Care

In 2021, an estimated 38.4 million people were living with diabetes, representing 11.6% of the U.S. population. This includes 38.1 million adults aged 18 years and older (14.7% of U.S. adults), of whom 8.7 million were unaware of their diagnosis. An additional 97.6 million adults aged 18 years and older (38.0% of the U.S. population) were living with prediabetes (6), and over 100 million adults 20 years and older (41.9% of U.S. adults) were living with obesity (7), placing them at high risk for developing type 2 diabetes. The prevalence of diabetes is increasing among children and adolescents as well. In 2017, the year for which the most recent data are available, the prevalence of type 1 diabetes was 2.15 per 10,000 individuals aged 19 years and under, and the prevalence of type 2 diabetes was 0.67 per 1,000 individuals 10–19 years of age (prevalence among those under 10 years old was too low to be reported) (8). Obesity prevalence has also increased significantly among children and adolescents in recent years, affecting approximately 14.7 million individuals between 2 and 19 years (19.7% of all U.S. children and adolescents) (9).

Most concerning is the recent rise in the rates of diabetes complications among adults with diabetes across the U.S. observed since the early 2010s, particularly for kidney failure, nontraumatic lower-extremity amputations, stroke, heart failure, and hyperglycemic crises, while rates of myocardial infarction have plateaued, reversing decades of earlier improvements (10). The proportion of people with diabetes who achieve recommended A1C, blood pressure, and cholesterol levels has fluctuated over the years, with some improvement over time (11,12). However, in 2015–2018 (the time frame for which the most recent population-level data are available), just 50.5% of U.S. community-dwelling adults with diabetes achieved A1C <7%, and 75.4% achieved A1C <8% (12). The goal blood pressure of <130/80 mmHg was achieved by just 47.7%, non-HDL cholesterol <130 mg/dL was achieved by 55.7%, and all three risk factors were treated to goal in just 22.2% (12). Importantly, many people who did not attain A1C, blood pressure, and lipid goals were not receiving any or adequate pharmacotherapy for glycemia, hypertension, and dyslipidemia management, respectively (i.e., were undertreated), which underscores the urgent need for care delivery systems and structural facilitators (i.e., health and public health policies and payment models) that enable timely and equitable delivery of evidence-based care and address diabetes prevention and treatment in communities (12).

Many segments of the population face additional and unique challenges to diabetes management, resulting in higher rates of diabetes complications, morbidity, and mortality. These include children, adolescents, and young adults; individuals with complex health needs, financial or other social hardships, and/or limited English proficiency; and individuals in groups and communities that have been historically marginalized (6,10,12,13). The prevalence of diabetes among adults is higher among non-Hispanic Black individuals (17.4%), non-Hispanic Asian individuals (16.7%), and Hispanic individuals (15.5%) than among White individuals (13.6%). Diabetes prevalence is also higher among adults with less than a high school education (13.1%) than among those with more than a high school education (6.9%), higher

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among those from families living below the federal poverty level (13.1%) than among those at 500% of federal poverty level of higher (5.1%), and higher among those living in nonmetropolitan areas (9.5%) than among those living in metropolitan areas (8.1%) (6). There is also substantial county-level variation in diabetes complications, mortality, and disparities in complications and mortality experienced by minoritized racial and ethnic groups across the U.S. (13), underscoring the impacts of broader structural factors on health among historically marginalized populations.

Gaps and disparities in diabetes management and outcomes are also prevalent among children, adolescents, and young adults with diabetes in the U.S. Improving diabetes management early in life is essential for preventing adverse health outcomes and disability and improving quality of life both in the short- and long-term. There are important and nuanced considerations in treating diabetes and supporting children, adolescents, and young adults with diabetes that are distinct from those of older adults (14). Data from SEARCH for Diabetes in Youth (SEARCH), a population-based registry network of five centers across five U.S. states, showed that in 2014–2019, mean A1C was 9.1% (76 mmol/mol) among youth and young adults with type 1 diabetes and 8.9% (74 mmol/mol) in youth and young adults with type 2 diabetes; these values increased from 8.5% (70 mmol/mol) and 8.4% (68 mmol/mol), respectively, in 2002–2007 (15). Youth and young adults with type 1 diabetes who are non-Hispanic Black or Native American (compared with those who are non-Hispanic White), younger, not treated with insulin pump therapy, and from households with low annual income have significantly higher A1C levels (15). Data from a network of academic pediatric and adult centers across the U.S. revealed that between 2016 and 2018, mean A1C was 8.1% (65 mmol/mol) among children with type 1 diabetes at 5 years of age and 9.3% (78 mmol/mol) among children 15–18 years of age. Only 17% of youth under 18 years of age with type 1 diabetes achieved the recommended A1C goal of <7.5% (<58 mmol/mol), and A1C levels for non-Hispanic Black youth were higher than those for non-Hispanic or Hispanic White youth, a disparity that persisted after adjustment for socioeconomic status (16).

Diabetes and its associated health complications pose a significant financial hardship to individuals and society. It is estimated that the annual cost of diagnosed diabetes in the U.S. in 2022 was ≥$413 billion, including ≥$307 billion in direct health care costs and ≥$106 billion in reduced productivity (17). After adjusting for inflation, the economic costs of diabetes increased by 7% between 2017 and 2022 and by 35% between 2012 and 2022 (17). This is attributed to both the increased prevalence of diabetes and the higher cost per person with diabetes. People living with diabetes also face individual financial hardship, resulting in rationing or not taking prescribed medications, delaying care, and foregoing or rationing other expenditures, among other behaviors, and ultimately correlating with higher A1C, greater diabetes distress, and depressive symptoms (18).

The growing gaps in diabetes care quality and outcomes, the high and rising costs of diabetes care across the U.S., and the disparities experienced by individuals from racial and ethnic minoritized backgrounds and those facing socioeconomic barriers to care call for urgent, substantial, and multisectoral system-level improvements to care delivery (19).

Evidence-Based Care Delivery and Payment Models to Improve Population Health

A major barrier to comprehensive high-quality diabetes care is a delivery system that is often fragmented, lacks clinical information capabilities, is not appropriately incentivized and funded, does not adequately engage people with diabetes and the communities where they live, and is poorly designed for the coordinated and longitudinal delivery of chronic care (20). Several models have been demonstrated to improve aspects of diabetes care delivery and health outcomes.

The Chronic Care Model (CCM) is a commonly used framework for describing diabetes care programs (21). It includes six core elements to optimize the care of people with chronic disease:

1. Delivery system design (moving from a reactive to a proactive care delivery system where planned visits are coordinated through a team-based approach)
2. Self-management support
3. Decision support, particularly at the point of care during a clinical encounter (basing care on evidence-based, effective care guidelines)
4. Clinical information systems (using registries that can provide person-specific and population-based support to the care team)
5. Community resources and policies (identifying or developing resources to support healthy lifestyles)
6. Health systems (to create a quality-oriented culture)

Randomized controlled trials of CCM interventions have shown that while interventions vary, programs that include core components of the CCM decrease A1C (mean difference [MD] −0.21% [95% CI −0.30 to −0.13], $P < 0.001$ compared with usual care), with greater improvements seen among adults with higher baseline A1C and with interventions that include four or more CCM elements (22). CCM-aligned programs also improved blood pressure levels and processes of diabetes care (e.g., screening for complications of diabetes), though there was no impact on cholesterol levels, tobacco use, or weight (23). Multiple studies have examined individual components of the CCM with respect to diabetes management and have found inconsistent levels of benefit with case management, team-based care, use of electronic patient registries, clinician education, clinician and patient reminders, and patient education and promotion of individual self-management (24). The inconsistencies in findings may be driven by heterogeneity of interventions, settings, and evaluation strategies as well as structural barriers to program utilization and needs of communities served.

Collaborative, interprofessional teams, which can bring together multiple disciplines within the health care system, payors, and community partners, are best suited to provide care for people with chronic conditions such as diabetes and to facilitate individuals' self-management (**Table 1.1**) (25–27). The care team, which centers around the person with diabetes, should avoid therapeutic inertia and prioritize timely and appropriate modification of behavior change interventions (nutrition and physical activity), pharmacologic therapy, technology use, and/or social and financial support systems for individuals who have not achieved recommended individualized metabolic goals or are experiencing high burden of treatment. Engagement of an endocrinology team with expertise in diabetes management is important in situations when the complexity of diabetes

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Table 1.1—Considerations for engaging interprofessional members of a comprehensive, person-centered diabetes care team to identify and meet the needs of people with diabetes across the life span

Subpopulation of a person with diabetes	Team members to engage in care	Unique care considerations
All adults with diabetes	Primary care clinician, CDCES, RDN, and other specialists as available and appropriate to treat comorbidities (**Table 4.1**)	Assess for and address social determinants of health.
Adults treated with intensive insulin therapy, including multiple daily injections of insulin and insulin pump therapy	Clinicians and other health care team members experienced in advanced diabetes management, including technology use	
All children and adolescents with diabetes	Primary care clinician, pediatric endocrinologist, CDCES, RDN, other specialists as available and appropriate to treat comorbidities (**Table 14.1**), daycare or school nurse or other professional, behavioral health professional (as needed), and parent(s) or caregiver(s)	Assess for and address social determinants of health and barriers to safety, well-being, and academic performance in school. Engage professionals within the school and extracurricular/after-school activities to ensure safe diabetes management. An individualized diabetes medical management plan should be developed in collaboration with school professionals and parent(s) or caregiver(s). Support gradual developmentally appropriate transfer of self-management from caregivers to the children and adolescents with diabetes.
Individuals with diabetes and diabetes-related complications or comorbidities	Specialist referrals as appropriate and available (e.g., behavioral health professional, cardiologist, endocrinologist, eye specialist, gastroenterologist or hepatologist, neurologist, nephrologist, obesity medicine specialist, or podiatrist), care coordinator/navigator or case manager, and clinical pharmacist (for those with polypharmacy or complex medication plans)	Screen for functional, cognitive, financial, and logistical barriers to self-management and evidence that self-care demands exceed capacity and available resources and support systems.
Individuals with social and/or structural barriers to care	Care coordinator/navigator, social services professional, insurance specialist/navigator, peer-to-peer support (as available), community health worker and/or community paramedic (as available), public health professional, and interpreter (as applicable)	Consider each person's psychosocial needs, available resources, and support systems.
Older adults	Geriatric medicine specialist, clinical pharmacist (for those with polypharmacy or complex medication plans), CDCES and/or RDN with expertise in the care for older adults, social services professional, case manager, community services provider, and physical and/or occupational therapist as available and appropriate based on functional status and independence	Consider the older adult's nutritional status, including ability to afford (financial barriers), acquire (accessibility), prepare (cooking), and consume (oral health) nutritious food. Assess for and address needs related to vision, hearing, dexterity, cognition, mobility, and other challenges.
Individuals in long-term care settings	Long-term care facility clinicians, nurses, other health care professionals, physical and occupational therapists, and RDN	Engage professionals within the long-term care facility to ensure safe and appropriate diabetes management.
Pregnant individuals with diabetes	Maternal-fetal medicine specialist or obstetrician experienced in the care of pregnant individuals with diabetes (particularly for individuals with type 1 diabetes or requiring intensive insulin therapy), CDCES, RDN, eye specialist (particularly for individuals with preexisting type 1 or type 2 diabetes), other specialists as appropriate, and lactation consultant as appropriate	Ensure appropriate postpartum follow-up and care, including transition from obstetric care to established primary care.
Individuals with behavioral health conditions	Behavioral health professional, care coordinator/navigator, and social services professional as age and situation appropriate	Use age- and situation-appropriate screening protocols for general and diabetes-related psychosocial concerns.

CDCES, certified diabetes care and education specialist; RDN, registered dietitian nutritionist.

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treatment exceeds the capacity of the primary care team to effectively manage, including in the setting of high clinical complexity and multimorbidity and difficult-to-manage hyperglycemia and/or hypoglycemia (28–32).

Initiatives such as the Patient-Centered Medical Home (PCMH) model can improve health outcomes by fostering comprehensive primary care and offering new opportunities for team-based chronic disease management (33–35). Accountable Care Organizations (ACOs), primary care–centered delivery and payment models, can support the implementation of the CCM and ultimately improve diabetes-related metrics in participating organizations (36). The Accountable Health Communities Model was introduced to support identifying and addressing health-related social needs to improve disease management and health outcomes (37); early evidence showed reduction in emergency department use among Medicare and Medicaid beneficiaries, but diabetes-specific metrics were not examined, and program effectiveness has been limited by scarcity of resources to meet identified health-related social needs (38). Alternative Payment Models (APMs) have had mixed effects on diabetes care delivery and outcomes, with higher-risk APMs (i.e., models with greater financial risk assumed by the provider, such as capitated payment models) generally associated with greater improvements in diabetes care processes than lower-risk APMs (39). Value-based payment models are hypothesized to better support the implementation and sustainability of innovative care delivery models seeking to improve population health (33,40), though evidence for currently available value-based insurance designs is limited (39).

Flexible Modalities of Care Delivery: Telehealth and Support for Behavioral Change and Well-being

Telehealth uses digital tools like video conferencing, mobile apps, and remote monitoring to deliver a range of health services remotely, including clinical care, education, and administrative support. Telemedicine, a subset of telehealth, focuses specifically on remote clinical care, supporting diagnosis, treatment, and consultations through real-time communication. Increased access to and effective use of telehealth services, alongside in-person care, can enhance timely access to diabetes care and DSMES services for individuals with diabetes (41–45).

Telehealth should be used to complement and not replace in-person visits for optimal glycemic management (46,47). A growing body of evidence supports the effectiveness of telehealth for glycemic management in people with diabetes. In a 2025 systematic umbrella review of 30 systematic reviews and meta-analyses, 28 reviews analyzed A1C and reported a significant reduction in A1C for people with diabetes (48). In the meta-analytic estimate, 16 of the 30 systematic reviews covering 681 unique trials showed mean reduction in A1C of 0.37% (95% CI −0.41% to −0.32%). Another 2025 systematic review and meta-analysis of 54 randomized controlled trials with people with type 2 diabetes and hypertension found a reduction in A1C of 0.45% (95% CI −0.90% to 0.00%) at 6 months and 0.18% (95% CI −0.41% to −0.05%) at 12 months compared with standard care, and systolic blood pressure was reduced by 5.63 mmHg (95% CI −9.13 to −2.13) at 6 months and 5.59 mmHg (95% CI −10.03 to −1.14) at 12 months compared with standard care (49). For people with type 1 diabetes, a 2023 systematic review and meta-analysis of 22 randomized controlled trials showed a reduction in A1C of 0.26% (95% CI −0.37% to −0.15%) (50). Furthermore, telehealth has been increasingly shown to help rural populations or those with limited physical access to health care in glycemic management as measured by A1C (51–54). In addition, evidence supports the effectiveness of telehealth in hypertension and dyslipidemia interventions (55). Interactive strategies that facilitate communication between health care professionals and people with diabetes, including the use of web-based portals or text messaging and those that incorporate medication adjustment, appear to be effective in improving outcomes (52,56).

Telehealth and other virtual environments can be used to offer diabetes self-management education (DSME), carbohydrate counting instruction, and clinical support while reducing geographic and transportation barriers for individuals living in underresourced areas or with disabilities (57). Telehealth resources can also have a role in improving diabetes management in children and adolescents with type 1 diabetes (58) and addressing SDOH in young adults with diabetes (59). Beyond pediatric populations, telehealth has demonstrated effectiveness in reducing disparities and improving glycemic outcomes in racially and ethnically diverse adults with type 2 diabetes. A systematic review and meta-analysis of nine telehealth randomized controlled trials aimed at improving A1C among Black and Hispanic adults with type 2 diabetes (>50% sample) found a −0.47% (95% CI −0.65% to −0.28%) change in A1C (60). While this systematic review and meta-analysis reported improved glycemic outcomes in marginalized populations, these successes typically depend on substantial investments in infrastructure (laptops, mobile phone access, internet access) and targeted support (digital literacy education, community health workers [CHWs]) that are rarely part of routine care (61). Without this infrastructure and support, attrition is high. Research on technology and inequities highlights that the digital divide is driven by financial (cost of data plans, high-speed internet), informational (numeracy, literacy, digital skills), and technical (interoperability gap, complex user interfaces) barriers (62). Therefore, effective telehealth design must consider accessibility, affordability, connectivity, trust, and contextual tailoring (62). Failure to do so risks exacerbating existing inequities.

Successful diabetes care also requires a systematic approach to supporting the behavior change efforts of people with diabetes. High-quality DSMES has been shown to improve a person's self-management, satisfaction, and glycemic outcomes (see section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes," for a detailed review of the evidence supporting DSMES). National DSMES standards call for an integrated approach that includes clinical content and skills, behavioral strategies (goal setting, problem-solving, etc.), and engagement with psychosocial concerns (63). Increasingly, such support is available through online or mobile platforms that can support user access and effectiveness. Findings from a 2025 systematic review and meta-analysis of 43 randomized controlled trials comparing DSME, diabetes self-management support, or DSMES delivered via mHealth interventions versus usual care or attention placebo control showed modest A1C benefits: DSME MD −0.3% (95% CI −0.6% to −0.1%; P = 0.002), diabetes self-management support MD −0.4% (95% CI −0.6% to −0.2%;

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$P < 0.001$), and DSMES MD −0.2% (95% CI −0.3% to −0.0%; $P < 0.001$) (64). These curricula should be tailored to the needs of their intended populations, including demographic characteristics, diabetes knowledge, emotional support, and cultural beliefs, and address the digital divide, i.e., access to the technology required for implementation (54,65,66).

Strategies for System-Level Improvement

Optimal diabetes management requires a systematic approach and coordinated team of health care professionals working in an environment where person-centered, high-quality care is a priority (23,67–69). While many diabetes care processes and access to technologies have improved nationally in the past decade, the overall quality of care for people with diabetes remains suboptimal (13). Efforts to increase the quality of diabetes care include providing care that is concordant with evidence-based guidelines (70), expanding the role of teams to implement more intensive disease management strategies (71), tracking medication-taking behavior (70), redesigning care processes (71), implementing electronic health record (EHR) population health tools (72), empowering and educating people with diabetes (73), reducing financial barriers (74), leveraging telehealth to improve access to care (51), assessing and addressing psychosocial issues (63,75), and engaging community resources and public policies that support healthy lifestyles (76). The National Diabetes Education Program maintains an online resource (cdc.gov/diabetes/php/toolkits/index.html) to help health care professionals design and implement more effective health care delivery systems for people with diabetes. Given the pluralistic needs of people with diabetes and the challenges they experience (complex insulin treatment plans, new technologies, changes in capacity for self-management, etc.) vary over the course of disease management and life span, engagement of an interprofessional team with complementary expertise is essential (25). Findings from a systematic review and meta-analysis of 35 team-based care interventions with adults with type 1 or type 2 diabetes showed a −0.5% (95% CI −0.7 to −0.3) change in A1C, −5.5 mmHg (95% CI −8.1 to −3.0) change in systolic blood pressure, −3.2 mmHg (95% CI −4.8 to −1.5) change in diastolic blood pressure,−8.0 mg/dL (95% CI −11.8 to −4.3) change in LDL cholesterol, and −7.4 mg/dL (95% CI −13.9 to −0.9) change in total cholesterol (25). Team-based care has also been shown to reduce atherosclerotic cardiovascular disease (ASCVD) risks. Results from the Center for Integrated and Novel Approaches in Vascular-Metabolic Disease (CINEMA) Program demonstrated improvements in ASCVD risk factors and evidence-based therapies with team-based care for adults with type 2 diabetes or prediabetes via reductions in body weight (−5.5 lb [−2.5 kg]), BMI (−0.9 kg/m^2), systolic (−3.6 mmHg) and diastolic (−1.2 mmHg) blood pressure, A1C (−0.5%), LDL (−9.0 mg/dL) and total cholesterol (−10.7 mg/dL), triglycerides (−13.5 mg/dL), and absolute 10-year ASCVD risk (−2.4%) (77).

A systematic review concluded that quality improvement (QI) initiatives can significantly improve outcomes for people with diabetes (24). As mandated by the Affordable Care Act, the Agency for Healthcare Research and Quality developed a National Quality Strategy based on three aims: improving the health of populations, improving overall quality and the personal experience of care, and reducing per capita cost (78). QI methods have been documented to improve diabetes device uptake, increase screening for psychosocial needs, and reduce inequities in access to diabetes technologies (79–82). Information and guidance specific to quality improvement and practice transformation for diabetes care are available from the National Institute of Diabetes and Digestive and Kidney Diseases guidance on diabetes care and quality (83).

A successful QI team should include a clinical champion, administrative leader, QI/data specialist, and an individual living with or affected by diabetes. Using patient registries and EHRs, health systems can evaluate the quality of diabetes care being delivered, benchmark metrics, and perform intervention cycles as part of QI strategies (19,72,83). QI can also be used as an effective strategy to support application of clinical practice recommendations by health care professionals.

The ADA recommends that health systems use proven QI methods to translate the Standards of Care recommendations into routine standards of practice.

In addition to QI approaches, other strategies that simultaneously improve the quality of care and potentially reduce costs are gaining momentum and include reimbursement structures that, in contrast to visit-based billing, reward the provision of appropriate and high-quality care to achieve metabolic goals (84), value-based payments, and incentives that accommodate personalized care goals (67,85). Alignment of payment models, health system care delivery models and resource availability, community-based organizations and supports, and regulatory policies is needed to fully support whole-person, comprehensive care for people with diabetes. See EVIDENCE-BASED CARE DELIVERY AND PAYMENT MODELS TO IMPROVE POPULATION HEALTH, above, for more information.

Access to Care

The Affordable Care Act and Medicaid expansion have increased access to care for many individuals with diabetes, emphasizing the protection of people with preexisting conditions, health promotion, and disease prevention (86). In 2024, population-level data for the U.S. civilian noninstitutionalized population included in the National Health Interview Survey (NHIS) showed that just 4.9% of children and adolescents (aged 17 years and younger), 11.1% of adults aged 18–64 years, and 0.6% of adults aged 65 years and older were uninsured (87). People with diabetes who have consistent private or public insurance coverage (i.e., are not uninsured) are more likely to meet quality indicators for diabetes care (86). However, even individuals with insurance coverage can experience financial barriers to care, particularly if enrolled in high-deductible health plans. In 2021, 28% of individuals with employer-sponsored health plans were enrolled in high-deductible health plans (88). Such plans are increasing in popularity; by 2023, 51% of private industry employees had the option to enroll in a high-deductible health plan, although only 36% had access to health savings accounts, which can offset some of the out-of-pocket costs incurred with high-deductible plans (89). Switching to a high-deductible health plan has been shown to increase financial hardship among people with diabetes (90), decrease and delay screening for retinopathy (91), decrease monitoring of blood pressure and A1C (91), and increase the risks of experiencing both acute (severe hypoglycemia, hyperglycemic crises) (92) and chronic (myocardial infarction, stroke, hospitalization for heart

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failure, kidney failure, lower-extremity complications, proliferative retinopathy, and blindness) (93) diabetes complications. However, health plan adoption of preventive drug lists that reduce or eliminate out-of-pocket cost sharing for a wide range of medications used to treat chronic conditions like diabetes reduces out-of-pocket expenditures for people with diabetes and narrows gaps in diabetes care (94). Insurance coverage and formulary design influence treatment decisions; it is essential that payors cover evidence-based diabetes care with minimal cost sharing by people with diabetes. Health care teams should also discuss insurance coverage and financial barriers to care with all individuals with diabetes and pursue therapeutic strategies that minimize financial hardship (74).

Access to primary and specialty care is essential for people with diabetes. While most adults with diabetes have access to a primary care clinician—a 2016 nationally representative population-based study found that 88% of adults with diabetes saw a primary care clinician in the prior year (95)—fewer have access to specialty endocrinology/diabetes care (96). A study of Medicare beneficiaries found that just 33% of older adults with type 1 diabetes, 14% of adults with type 2 diabetes and history of severe hypoglycemia, and 9% of other adults with type 2 diabetes saw an endocrinologist in 2019 (96). Racial and ethnic minoritized individuals, those with low income, those living in rural areas, and those residing in a long-term care facility were less likely to receive endocrinology care. Improving health outcomes for people with diabetes will therefore require improving availability of and access to primary and specialty services, and interdisciplinary care from the full diabetes care team, to meet the full range of their health care needs (**Table 1.1**).

Cost Considerations

The cost of diabetes medications and devices is an ongoing barrier to achieving glycemic goals and preventing diabetes complications. Medication underuse due to cost has been termed "cost-related medication nonadherence" (here referred to as cost-related barriers to medication use). Based on a national survey conducted in 2021, 18.6% of U.S. adults with type 1 diabetes and 15.8% of adults with insulin-treated type 2 diabetes reported rationing (i.e., skipping, taking less, and/or delaying) their insulin to save money (97). The ADA Insulin Access and Affordability Working Group has recommended system-level approaches to address this issue, including concepts such as cost-sharing for insured people with diabetes based on the lowest price available, a list price for insulins that closely reflects the net price, and health plans that ensure people with diabetes can access insulin without undue administrative burden or excessive cost (98). In 2021, the Centers for Medicare & Medicaid Services (CMS) launched the Part D Senior Savings Model (99), which required capped monthly out-of-pocket payments for covered insulins at $35 per insulin per month. In 2022, 43% of standalone Part D plan enrollees and 60% of Medicare Advantage Part D plan enrollees participated in the Senior Savings Model (99). Most recently, the Inflation Reduction Act of 2022 capped out-of-pocket payments for insulin at $35 per insulin per month for all Medicare beneficiaries. A patchwork of solutions has also been introduced for individuals with commercial insurance and those without health insurance. Over the past 5 years, 25 states and the District of Columbia have capped out-of-pocket expenditures for insulin in select state-regulated commercial health plans (100), which has resulted in modest decreases in insulin out-of-pocket spending for eligible enrollees (101). Between 2023 and 2024, three major insulin manufacturers similarly lowered the price of insulin to $35 per month in select circumstances (102). These programs may help reduce the financial hardship of diabetes management, though many are challenging to navigate, not all people with diabetes can benefit, and costs for insulin delivery and glucose monitoring remain high. Thus, all people with diabetes should be screened for financial hardship of treatment, cost-related barriers to medication use, and rationing of other essential services due to medical costs (103).

The cost of medications (not only insulin) influences prescribing patterns and medication use because of the financial strain on the person with diabetes and the lack of secondary payor support (public and private insurance) for effective approved glucose-lowering, cardiovascular and kidney disease risk–reducing, and obesity management therapies. There is robust evidence of gaps in the use of evidence-based therapies among individuals from racial and ethnic minoritized backgrounds, those with lower income levels, those living in rural areas, and those with limited insurance coverage (11,104–112). Identifying and addressing financial barriers should be a focus of treatment goals and clinical decisions (104). (See TAILORING TREATMENT FOR SOCIAL CONTEXT.)

TAILORING TREATMENT FOR SOCIAL CONTEXT

Recommendations

1.6 Health systems should assess and address gaps in diabetes care and health outcomes (e.g., by stratifying clinical quality data by factors such as insurance status, race, ethnicity, preferred language for health care discussions, disability, and social determinants of health). **B**

1.7 During clinical encounters, assess for social determinants of health, including food insecurity, **A** housing insecurity, financial barriers, health insurance and health care access, environmental and neighborhood factors, and social capital/social community support, **B** to inform treatment decisions, with referral to appropriate local community resources.

1.8 Provide people with diabetes additional self-management support from lay health coaches, navigators, or community health workers when available. **A** Digital self-management tools or coaches may be considered as appropriate. **B**

1.9 Consider the involvement of community health workers to support management of diabetes and cardiovascular and kidney risk factors, especially in underserved communities and health care systems. **B**

Health inequities related to diabetes and its complications are well documented, are heavily influenced by SDOH, and have been associated with greater risk for developing diabetes, higher disease prevalence, and worse diabetes-related outcomes (4). Greater exposure to adverse SDOH over the life course results in poor health (113). Interventions to address SDOH can improve diabetes-related outcomes (114,115). Using clinical quality data to identify inequities

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and opportunities for improvement is valuable for health care professionals, health systems, payors, policymakers, and people with diabetes (116). The Joint Commission requires that all accredited organizations in its ambulatory health care, behavioral health care and human services, critical access hospital, and hospital accreditation programs collect race and ethnicity information and implement specific steps to reduce health care disparities. The Joint Commission specifically requires that organizations designate an individual or individuals to lead efforts to reduce health care disparities, assess health-related social needs and provide information on community resources to meet these needs, identify health care disparities by stratifying quality and safety data using sociodemographic characteristics, develop an action plan to address health care disparities, work to actively reduce health care disparities, and inform key stakeholders about progress to reduce health care disparities (117). The CMS Framework for Health Equity similarly prioritizes collection, reporting, and analysis of standardized individual-level demographic (including race, ethnicity, language, gender identity, sex, sexual orientation, and disability status) and SDOH data as well as assessing for and addressing disparities through improved access to culturally tailored services, team-based care, and community resources (118). Quality measures assessing SDOH screening and intervention have been introduced by the National Committee for Quality Assurance (focused on food, housing, and transportation insecurity) (113) and CMS (focused on food, housing, and transportation insecurity, utility difficulties, and interpersonal safety) (119).

Other than the SDOH, there are many contributors to inequities, including implicit and explicit bias, discriminatory policies and procedures, institutional practices, and both current and historical structural factors that affect health and well-being for individuals and communities (120–122). The ADA recognizes the association between interpersonal, social, economic, and environmental factors and the prevention and treatment of diabetes and has issued a call for research that seeks to better understand how social determinants influence behaviors and how the relationships between these variables can be modified for enhancing the prevention and management of diabetes (114). While a comprehensive strategy to reduce diabetes-related health disparities in populations is yet to be formally studied, general recommendations from other chronic disease management and prevention models can be drawn upon to inform system-level strategies in diabetes (62). For example, the National Academy of Medicine has published a framework for educating health care professionals on the importance of SDOH (123). Furthermore, there are resources available for the inclusion of standardized sociodemographic variables in EHRs to facilitate the assessment of health disparities and the impact of interventions designed to reduce those disparities (78,124,125).

SDOH are not consistently recognized and often go undiscussed—and are ultimately not addressed—during clinical encounters (126). Even financial barriers to care and health, which directly affect the ability to implement prescribed care plans, are rarely discussed. Among people with chronic illnesses, two-thirds of those who reported not taking medications as prescribed due to cost-related barriers never shared this information with their physician (123). A study using data from the NHIS found that half of adults with diabetes reported financial stress and about 20% reported food insecurity (126). Studies of both type 1 diabetes and type 2 diabetes have noted an association of one or more adverse SDOH with health care utilization and poor diabetes outcomes among individuals with diabetes (123,127). It is therefore important for people with diabetes to be screened for SDOH during clinical encounters and be referred to appropriate clinical and community resources to address these needs (**Table 1.1**). Furthermore, health systems may benefit from compiling an inventory of local resources to facilitate referrals at the point of care, augmenting as necessary existing tools such as findhelp.org, Unite Us, 211 (United Way), and others. Policies and payment models that support addressing SDOH, both within and outside the health care setting, are needed to ensure that these efforts are both feasible and sustainable. One example of a statewide payment model that incentivizes value-based care, addressing SDOH and funding community-based health care professionals, is the Maryland Total Cost of Care Model, although it is currently limited by a narrow focus on preventing diabetes and does not consider diabetes care quality or health outcomes in people with diabetes (116,128).

A population in which such issues must be particularly considered is older adults, for whom social difficulties may further impair quality of life and increase the risk of functional dependency (129) (see section 13, "Older Adults," for a detailed discussion of social considerations in older adults).

Creating system-level mechanisms to screen for SDOH may help overcome structural barriers and communication gaps between people with diabetes and health care professionals (126,130). A number of studies have proven the effectiveness of identifying SDOH by using validated screening tools (131). In addition, brief, validated screening tools for some SDOH exist and could facilitate discussion around factors that significantly impact treatment during clinical encounters.

Food Insecurity

Food insecurity is a household-level economic and social condition of limited or uncertain access to adequate food (132). In 2022, almost 13% of Americans were food insecure (132), and food insecurity is associated with increased risk of type 2 diabetes and higher-than-recommended glycemia (133–135). The rate is disproportionately higher among some groups that have been historically marginalized, low-income households, and households headed by single mothers. Additionally, those facing food insecurity have lower engagement in self-care behaviors and medication use, have higher rates of depression and diabetes distress, and have worse glycemic management compared with individuals who are food secure (133,134). Older adults with food insecurity are more likely to have emergency department visits and hospitalizations compared with older adults who do not report food insecurity (136). The risk for food insecurity can be assessed with a validated two-item screening tool that includes the following statements: *1*) "Within the past 12 months, we worried whether our food would run out before we got money to buy more" and *2*) "Within the past 12 months the food we bought just didn't last, and we didn't have money to get more" (137). Interventions such as food prescription programs are considered promising to address food insecurity by integrating community resources into primary care settings and directly dealing

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with food deserts in underserved communities (135).

In those with diabetes and food insecurity, the priority is mitigating the increased risk for severe hyperglycemia and hypoglycemia (138,139). The reasons for the increased risk of hyperglycemia can include the consumption of inexpensive carbohydrate-rich processed foods, binge eating, financial constraints to filling diabetes medication prescriptions, anxiety and depression, and poor sleep, all contributing to hyperglycemia and poor diabetes self-care behaviors. Hypoglycemia can occur as a result of inadequate or inconsistent carbohydrate consumption following the administration of a sulfonylurea or insulin. Health care professionals should consider these factors when making treatment decisions for people with food insecurity and seek local resources to help people with diabetes and their family members obtain appropriate food more regularly (140).

Housing Insecurity

Housing insecurity has been shown to be directly associated with a person's ability to maintain appropriate diabetes self-management (141). Housing insecurity often coexists with other barriers that challenge diabetes self-management. Food insecurity, lack of health insurance, cognitive impairment, behavioral health concerns, and low literacy and numeracy skills are also factors (140). The prevalence of diabetes among people experiencing housing insecurity is estimated to be around 8% in the U.S. (142). Additionally, people with diabetes and housing insecurity need secure places to keep their diabetes medications and supplies as well as refrigerator access to safely store insulin. The risk for housing insecurity can be ascertained using a brief risk assessment tool developed and validated for use among veterans (143). Given the potential challenges, health care professionals who care for housing-insecure individuals should be familiar with resources to support these individuals or have access to social workers who can facilitate stable housing as a way to improve diabetes care (144).

Refugee Populations and Migrant and Seasonal Agricultural Workers

Refugee status, like having a diabetes diagnosis, is an independent risk factor for cardiovascular disease (145). In areas experiencing humanitarian crises, refugees are at greater risk for obstacles to optimal chronic disease management, but unfortunately there are few quality investigations into diabetes care and its outcomes among refugees. There have been efforts to develop models of care specifically aimed at improving the health of refugee populations, but more work is needed to demonstrate effectiveness of those care models and approaches (146).

Migrant and seasonal agricultural workers likely have a higher risk of type 2 diabetes than the general population. While migrant farmworker–specific data are lacking, most agricultural workers in the U.S. are Latino, a population with a high rate of type 2 diabetes. In addition, living in severe poverty brings with it food insecurity, high chronic stress, and an increased risk of diabetes; there is also an association between the exposure to certain pesticides and the incidence of diabetes (147).

Data from the Department of Labor indicate that there are approximately 2.18 million agricultural workers in the U.S. (148). These agricultural workers often travel throughout the country seasonally (149). In 2022, 175 health centers across the U.S. reported providing care to 843,071 adult migrant farmworkers, and 91,839 had encounters for diabetes (147). In a 2023 report on the National Agricultural Workers Survey, age-adjusted self-reported diabetes prevalence was 13.51% (95% CI 10.0–17.1) among migrant farmworkers and 10.8% (95% CI 9.0–12.6) among nonmigrant farmworkers (147).

Migrant farmworkers and other agricultural workers encounter numerous and overlapping barriers to receiving care. Migration, which might occur as frequently as every few weeks for some, disrupts care. Common barriers to adequate diabetes care include those related to cost, culture, language, literacy, transportation, geographic distance, food access, long work hours, unfamiliarity with new communities, the complexity of the U.S. health care system, and limited access to various other resources like medications and DSMES (149). Without regular care, farmworkers with diabetes can experience severe and often expensive complications that incur morbidity and mortality and affect quality of life. Nontraditional care delivery models, including mobile integrated health and telehealth, should be leveraged to improve access to care.

Health care professionals need to be attuned to the working and living conditions of people with diabetes. For example, if a farmworker with diabetes presents for care, appropriate referrals should be initiated to social workers and community resources, as available, to assist with removing barriers to care.

Language Barriers

Health systems and health care professionals caring for those with limited English proficiency should develop or offer educational programs and materials in culturally appropriate languages. Professional language assistance (i.e., interpreters) should be provided to individuals with limited English proficiency and/or other communication needs at no cost to them (150). Use of untrained interpreters, including family members, should be avoided when possible, as this can result in confusing or inaccurate conveyance of information. Accompanying written materials should be in the language appropriate for the individual being supported and at a reading level that is not overly complicated—typically this is defined as a sixth-grade reading level. Numerous medical translation tools are available and show promise in addressing language barriers, although further research is needed to establish their efficacy in diabetes care (151).

The National Standards for Culturally and Linguistically Appropriate Services in Health and Health Care (National CLAS Standards) provide guidance on how health care professionals can reduce language barriers by improving their cultural competency, addressing health literacy, and ensuring communication with professional language assistance (150). In addition, the National CLAS Standards website offers several resources and materials that can be used to improve the quality of care delivery to individuals with limited English proficiency.

Health Literacy and Numeracy

Health literacy is the degree to which individuals can obtain, process, and understand basic health information and services needed to make appropriate decisions (152,153). Health literacy is strongly associated with individuals engaging in complex disease management and self-care (154). Approximately 9 out of 10 American adults are estimated to have limited or low health literacy (152,155). Health care professionals and diabetes care and education specialists should provide easy-to-understand

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information and reduce unnecessary complexity when developing care plans. Interventions addressing low health literacy in populations with diabetes demonstrate improvements in A1C, diabetes knowledge, and self-management behaviors (156). However, more research is needed to establish the most effective strategies for enhancing retention and application of diabetes knowledge among various populations of people with diabetes (154,157).

Health numeracy is also essential in diabetes prevention and management. Health numeracy requires primary numeric skills, applied health numeracy, and interpretive health numeracy, which is especially important for people using diabetes technologies like insulin pumps (158). People with prediabetes or diabetes often need to perform numeric tasks, such as interpreting glucose readings, decoding food labels, and understanding medication doses, timings, and refills, to make treatment decisions. Thus, both health literacy and numeracy are necessary for enabling effective communication between people with diabetes and health care professionals, arriving at a treatment plan, and making diabetes self-management task decisions. If people with diabetes appear not to understand concepts associated with treatment decisions, both can be assessed using standardized screening measures (159). Adjunctive education and support may be indicated if limited health literacy and numeracy are barriers to optimal care decisions (75).

Social and Community Support

Social support, which comprises community and personal network instrumental support, promotes better health, whereas lack of social support is associated with poorer health outcomes in individuals with diabetes (114). Of particular concern are the SDOH, including, among others, racism and discrimination (160). These factors are rarely addressed in routine clinical practice but may be drivers of adverse health outcomes and lower engagement in beneficial self-care behaviors and medication use. Identifying and leveraging community resources are core components of chronic care management (21,161).

Various organizations, including the American Medical Association and the Agency for Healthcare Research and Quality, among others, promote the translation of clinical recommendations for nutrition and physical activity into practice through health care community linkages and community-based care delivery (162). CHWs (160), community paramedics (163,164), peer supporters (162,165), and lay leaders (166) may assist in the delivery of DSMES services (124,167), particularly in underserved communities. The American Public Health Association defines a CHW as a "frontline public health worker who is a trusted member of and/or has an unusually close understanding of the community served" (168). CHWs can be part of an evidence-based strategy to improve the management of diabetes and cardiovascular risk factors in underserved communities and health care systems (169). The CHW scope of practice in areas such as outreach and communication, advocacy, social support, basic health education, referrals to community clinics, and other services has successfully provided social and primary preventive services to underserved populations in rural and hard-to-reach communities. Even though CHWs' core competencies are not clinical in nature, in some circumstances, clinicians may delegate limited clinical tasks to CHWs. If such is the case, these tasks must always be performed under the direct supervision of the delegating health professional and following state health care laws and statutes (170,171). Community paramedics are advanced paramedics with training in chronic disease monitoring and education, medication management, care coordination, and SDOH in addition to their emergency medical services expertise. While their scope of practice varies across states, community paramedics can engage and support people living with diabetes under the direction of a medical director by delivering diabetes education, assisting with medication management, performing health assessments and wound care, and connecting people with diabetes and care partners with clinical and community resources (163,164).

SUMMARY

Improving individual and population health for people with and at risk for diabetes requires engagement of and collaboration between people with diabetes and their caregivers, interprofessional health care teams, health systems, community partners, payors, policymakers, and public health agencies. This section provides guidance to facilitate implementation of evidence-based diabetes care recommendations that are discussed in the Standards of Care with the goal of improving health, eliminating health disparities, and reducing the impact of diabetes and its complications on individuals and society.

References

1. Silberberg M, Martinez-Bianchi V, Lyn MJ. What is population health? Prim Care 2019;46:475–484
2. Bodenheimer T, Sinsky C. From triple to quadruple aim: care of the patient requires care of the provider. Ann Fam Med 2014;12:573–576
3. Institute of Medicine (US) Committee on Quality of Health Care in America. *Crossing the Quality Chasm: A New Health System for the 21st Century.* Washington, DC, National Academies Press, 2001
4. Hill-Briggs F, Adler NE, Berkowitz SA, et al. Social determinants of health and diabetes: a scientific review. Diabetes Care 2020;44:258–279
5. Haire-Joshu D, Hill-Briggs F. The next generation of diabetes translation: a path to health equity. Annu Rev Public Health 2019;40:391–410
6. Centers for Disease Control and Prevention. National Diabetes Statistics Report. Accessed 4 August 2025. Available from https://www.cdc.gov/diabetes/php/data-research/index.html
7. Centers for Disease Control and Prevention. Adult Obesity Facts. Accessed 4 August 2025. Available from https://www.cdc.gov/obesity/adult-obesity-facts/index.html
8. Lawrence JM, Divers J, Isom S, et al.; SEARCH for Diabetes in Youth Study Group. Trends in prevalence of type 1 and type 2 diabetes in children and adolescents in the US, 2001–2017. JAMA 2021;326:717–727
9. Centers for Disease Control and Prevention. Childhood Obesity Facts. Accessed 4 August 2025. Available from https://www.cdc.gov/obesity/childhood-obesity-facts/childhood-obesity-facts.html
10. Saelee R, Bullard KM, Hora IA, et al. Trends and inequalities in diabetes-related complications among U.S. Adults, 2000–2020. Diabetes Care 2025;48:18–28
11. Ebekozien O, Mungmode A, Sanchez J, et al. Longitudinal trends in glycemic outcomes and technology use for over 48,000 people with type 1 diabetes (2016-2022) from the T1D Exchange Quality Improvement Collaborative. Diabetes Technol Ther 2023;25:765–773
12. Fang M, Wang D, Coresh J, Selvin E. Trends in diabetes treatment and control in U.S. adults, 1999–2018. N Engl J Med 2021;384:2219–2228
13. Nassereldine H, Li Z, Compton K, et al. The burden of diabetes mortality by county, race, and ethnicity in the U.S., 2000–2019. Diabetes Care 2025;48:546–555
14. Chiang JL, Maahs DM, Garvey KC, et al. Type 1 diabetes in children and adolescents: a position statement by the American Diabetes Association. Diabetes Care 2018;41:2026–2044
15. Malik FS, Sauder KA, Isom S, et al.; SEARCH for Diabetes in Youth Study. Trends in glycemic control among youth and young adults with diabetes: the SEARCH for Diabetes in Youth Study. Diabetes Care 2022;45:285–294
16. Foster NC, Beck RW, Miller KM, et al. State of type 1 diabetes management and outcomes from the T1D Exchange in 2016–2018. Diabetes Technol Ther 2019;21:66–72

Downloaded from http://diabetesjournals.org/care/article-pdf/49/Supplement_1/S13/848832/dc26s001.pdf by guest on 18 December 2025

17. Parker ED, Lin J, Mahoney T, et al. Economic costs of diabetes in the U.S. in 2022. Diabetes Care 2024;47:26–43
18. Patel MR, Zhang G, Heisler M, et al. Measurement and validation of the Comprehensive Score for Financial Toxicity (COST) in a population with diabetes. Diabetes Care 2022;45:2535–2543
19. Prahalad P, Hardison H, Odugbesan O, et al.; T1D Exchange Quality Improvement Collaborative. Benchmarking diabetes technology use among 21 U.S. pediatric diabetes centers. Clin Diabetes 2024; 42:27–33
20. Chehal PK, Selvin E, DeVoe JE, Mangione CM, Ali MK. Diabetes and the fragmented state of US health care and policy. Health Aff (Millwood) 2022;41:939–946
21. Stellefson M, Dipnarine K, Stopka C. The chronic care model and diabetes management in US primary care settings: a systematic review. Prev Chronic Dis 2013;10:E26
22. Goh LH, Siah CJR, Tam WWS, Tai ES, Young DYL. Effectiveness of the chronic care model for adults with type 2 diabetes in primary care: a systematic review and meta-analysis. Syst Rev 2022;11:273
23. Tricco AC, Ivers NM, Grimshaw JM, et al. Effectiveness of quality improvement strategies on the management of diabetes: a systematic review and meta-analysis. Lancet 2012;379: 2252–2261
24. Konnyu KJ, Yogasingam S, Lépine J, et al. Quality improvement strategies for diabetes care: effects on outcomes for adults living with diabetes. Cochrane Database Syst Rev 2023;5: CD014513
25. Levengood TW, Peng Y, Xiong KZ, et al.; Community Preventive Services Task Force. Team-based care to improve diabetes management: a community guide meta-analysis. Am J Prev Med 2019;57:e17–e26
26. Lee JK, McCutcheon LRM, Fazel MT, Cooley JH, Slack MK. Assessment of interprofessional collaborative practices and outcomes in adults with diabetes and hypertension in primary care: a systematic review and meta-analysis. JAMA Netw Open 2021;4:e2036725
27. Oshman L, Bhomia N, Diez HL, et al. The Michigan Collaborative for Type 2 Diabetes (MCT2D): development and implementation of a statewide collaborative quality initiative. BMC Health Serv Res 2024;24:1254
28. Avnat E, Chodick G, Shalev V. Identifying profiles of patients with uncontrolled type 2 diabetes who would benefit from referral to an endocrinologist. Endocr Pract 2023;29:855–861
29. Setji TL, Page C, Pagidipati N, Goldstein BA. Differences in achieving Hba1c goals among patients seen by endocrinologists and primary care providers. Endocr Pract 2019;25:461–469
30. McCoy RG, Vandergrift JL, Gray B. Patient and physician factors driving the gaps in use of drugs with cardiovascular and kidney benefits by Medicare beneficiaries with type 2 diabetes treated by endocrinologists, nephrologists, and cardiologists: population-based cohort study. Diabetes Res Clin Pract 2025;221:112039
31. Phan T-L, Hossain J, Lawless S, Werk LN. Quarterly visits with glycated hemoglobin monitoring: the sweet spot for glycemic control in youth with type 1 diabetes. Diabetes Care 2014; 37:341–345
32. Booth GL, Shah BR, Austin PC, Hux JE, Luo J, Lok CE. Early specialist care for diabetes: who benefits most? A propensity score-matched cohort study. Diabet Med 2016;33:111–118
33. Adler-Milstein J, Linden A, Hollingsworth JM, Ryan AM. Association of primary care engagement in value-based reform programs with health services outcomes: participation and synergies. JAMA Health Forum 2022;3:e220005
34. Bojadzievski T, Gabbay RA. Patient-centered medical home and diabetes. Diabetes Care 2011; 34:1047–1053
35. McManus LS, Dominguez-Cancino KA, Stanek MK, et al. The patient-centered medical home as an intervention strategy for diabetes mellitus: a systematic review of the literature. Curr Diabetes Rev 2021;17:317–331
36. Fraze TK, Lewis VA, Tierney E, Colla CH. Quality of care improves for patients with diabetes in Medicare shared savings accountable care organizations: organizational characteristics associated with performance. Popul Health Manag 2018;21:401–408
37. Centers for Medicare & Medicaid Services. Accountable Health Communities Model. Accessed 8 August 2025. Available from https://www.cms.gov/priorities/innovation/innovation-models/ahcm
38. Centers for Medicare & Medicaid Services. Accountable Health Communities (AHC) Model Evaluation. Second Evaluation Report Accessed 8 August 2025. Available from https://www.cms.gov/priorities/innovation/data-and-reports/2023/ahc-second-eval-rpt
39. Wang S, Weyer G, Duru OK, Gabbay RA, Huang ES. Can alternative payment models and value-based insurance design alter the course of diabetes in the United States? Health Aff (Millwood) 2022;41:980–984
40. Gunter KE, Peek ME, Tanumihardjo JP, et al. Population health innovations and payment to address social needs among patients and communities with diabetes. Milbank Q 2021; 99:928–973
41. Katula JA, Dressler EV, Kittel CA, et al. Effects of a digital diabetes prevention program: an RCT. Am J Prev Med 2022;62:567–577
42. Eberle C, Stichling S. Clinical improvements by telemedicine interventions managing type 1 and type 2 diabetes: systematic meta-review. J Med Internet Res 2021;23:e23244
43. American Telemedicine Association. Telehealth: defining 21st Century Care. Accessed 8 August 2025. Available from https://www.americantelemed.org/resource/why-telemedicine/
44. Health IT. Telehealth StartUp and Resource Guide Version 1.1. Accessed 8 August 2025. Available from https://www.healthit.gov/sites/default/files/telehealthguide_final_0.pdf
45. American Medical Association. AMA Telehealth Quick Guide. Accessed 8 August 2025. Available from https://www.ama-assn.org/practice-management/digital/ama-telehealth-quick-guide
46. Zupa MF, Vimalananda VG, Rothenberger SD, et al. Patterns of telemedicine use and glycemic outcomes of endocrinology care for patients with type 2 diabetes. JAMA Netw Open 2023;6: e2346305
47. Mullur RS, Hsiao JS, Mueller K. Telemedicine in diabetes care. Am Fam Physician 2022;105: 281–288
48. Ravi S, Meyerowitz-Katz G, Yung C, et al. Effect of virtual care in type 2 diabetes management—a systematic umbrella review of systematic reviews and meta-analysis. BMC Health Serv Res 2025; 25:348
49. Mihevc M, Virtič Potočnik T, Zavrnik Č, Klemenc-Ketiš Z, Poplas Susič A, Petek Šter M. Managing cardiovascular risk factors with telemedicine in primary care: a systematic review and meta-analysis of patients with arterial hypertension and type 2 diabetes. Chronic Illn 2025;21:3–24
50. Udsen FW, Hangaard S, Bender C, et al. The effectiveness of telemedicine solutions in type 1 diabetes management: a systematic review and meta-analysis. J Diabetes Sci Technol 2023;17: 782–793
51. Kobe EA, Lewinski AA, Jeffreys AS, et al. Implementation of an intensive telehealth intervention for rural patients with clinic-refractory diabetes. J Gen Intern Med 2022;37:3080–3088
52. Heitkemper EM, Mamykina L, Travers J, Smaldone A. Do health information technology self-management interventions improve glycemic control in medically underserved adults with diabetes? A systematic review and meta-analysis. J Am Med Inform Assoc 2017;24:1024–1035
53. Jarvandi S, Roberson P, Greig J, Upendram S, Grion J. Effectiveness of diabetes education interventions in rural America: a systematic review. Health Educ Res 2023;38:286–305
54. Moschonis G, Siopis G, Jung J, et al.; DigiCare4You Consortium. Effectiveness, reach, uptake, and feasibility of digital health interventions for adults with type 2 diabetes: a systematic review and meta-analysis of randomised controlled trials. Lancet Digit Health 2023;5:e125–e143
55. Timpel P, Oswald S, Schwarz PEH, Harst L. Mapping the evidence on the effectiveness of telemedicine interventions in diabetes, dyslipidemia, and hypertension: an umbrella review of systematic reviews and meta-analyses. J Med Internet Res 2020;22:e16791
56. Faruque LI, Wiebe N, Ehteshami-Afshar A, et al.; Alberta Kidney Disease Network. Effect of telemedicine on glycated hemoglobin in diabetes: a systematic review and meta-analysis of randomized trials. CMAJ 2017;189:E341–E364
57. Li Y, Yang Y, Liu X, Zhang X, Li F. Effectiveness of telemedicine-delivered carbohydrate-counting interventions in patients with type 1 diabetes: systematic review and meta-analysis. J Med Internet Res 2025;27:e59579
58. Zhang K, Huang Q, Wang Q, et al. Telemedicine in improving glycemic control among children and adolescents with type 1 diabetes mellitus: systematic review and meta-analysis. J Med Internet Res 2024;26:e51538
59. Garcia JF, Fogel J, Reid M, Bisno DI, Raymond JK. Telehealth for young adults with diabetes: addressing social determinants of health. Diabetes Spectr 2021;34:357–362
60. Anderson A, O'Connell SS, Thomas C, Chimmanamada R. Telehealth interventions to improve diabetes management among Black and Hispanic patients: a systematic review and meta-analysis. J Racial Ethn Health Disparities 2022;9:2375–2386
61. Mayberry LS, Lyles CR, Oldenburg B, Osborn CY, Parks M, Peek ME. mHealth interventions for disadvantaged and vulnerable people with type 2 diabetes. Curr Diab Rep 2019;19:148
62. Ebekozien O, Fantasia K, Farrokhi F, Sabharwal A, Kerr D. Technology and health inequities in

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diabetes care: how do we widen access to underserved populations and utilize technology to improve outcomes for all? Diabetes Obes Metab 2024;26(Suppl. 1):3–13
63. Davis J, Fischl AH, Beck J, et al. 2022 National Standards for Diabetes Self-Management Education and Support. Diabetes Care 2022;45:484–494
64. Versluis A, Boels AM, Huijden MCG, Mijnsbergen MD, Rutten GEHM, Vos RC. Diabetes self-management education and support delivered by mobile health (mHealth) interventions for adults with type 2 diabetes—a systematic review and meta-analysis. Diabet Med 2025;42:e70002
65. Tanhapour M, Peimani M, Rostam Niakan Kalhori S, et al. The effect of personalized intelligent digital systems for self-care training on type II diabetes: a systematic review and meta-analysis of clinical trials. Acta Diabetol 2023; 60:1599–1631
66. Abu S, Llahana S. Factors influencing the uptake of culturally tailored diabetes self-management education and support programmes among ethnic minority patients with type 2 diabetes: a systematic review. Prim Care Diabetes 2025;19:103–110
67. TRIAD Study Group. Health systems, patients factors, and quality of care for diabetes: a synthesis of findings from the TRIAD study. Diabetes Care 2010;33:940–947
68. Schmittdiel JA, Gopalan A, Lin MW, Banerjee S, Chau CV, Adams AS. Population health management for diabetes: health care system-level approaches for improving quality and addressing disparities. Curr Diab Rep 2017;17:31
69. Peterson KA, Carlin CS, Solberg LI, Normington J, Lock EF. Care management processes important for high-quality diabetes care. Diabetes Care 2023; 46:1762–1769
70. Raebel MA, Schmittdiel J, Karter AJ, Konieczny JL, Steiner JF. Standardizing terminology and definitions of medication adherence and persistence in research employing electronic databases. Med Care 2013;51:S11–S21
71. Feifer C, Nemeth L, Nietert PJ, et al. Different paths to high-quality care: three archetypes of top-performing practice sites. Ann Fam Med 2007;5:233–241
72. Mungmode A, Noor N, Weinstock RS, et al. Making diabetes electronic medical record data actionable: promoting benchmarking and population health improvement using the T1D Exchange Quality Improvement Portal. Clin Diabetes 2022;41:45–55
73. Powell RE, Zaccardi F, Beebe C, et al. Strategies for overcoming therapeutic inertia in type 2 diabetes: a systematic review and meta-analysis. Diabetes Obes Metab 2021;23:2137–2154
74. Herges JR, Neumiller JJ, McCoy RG. Easing the financial burden of diabetes management: a guide for patients and primary care clinicians. Clin Diabetes 2021;39:427–436
75. Young-Hyman D, de Groot M, Hill-Briggs F, Gonzalez JS, Hood K, Peyrot M. Psychosocial care for people with diabetes: a position statement of the American Diabetes Association. Diabetes Care 2016;39:2126–2140
76. Pullen-Smith B, Carter-Edwards L, Leathers KH. Community health ambassadors: a model for engaging community leaders to promote better health in North Carolina. J Public Health Manag Pract 2008;14(Suppl.):S73–S81
77. Neeland IJ, Arafah A, Bourges-Sevenier B, et al. Second-year results from CINEMA: a novel, patient-centered, team-based intervention for patients with type 2 diabetes or prediabetes at high cardiovascular risk. Am J Prev Cardiol 2024; 17:100630
78. Kahkoska AR, Busby-Whitehead J, Jonsson Funk M, et al. Receipt of diabetes specialty care and management services by older adults with diabetes in the U.S., 2015–2019: an analysis of Medicare fee-for-service claims. Diabetes Care 2024;47:1181–1185
79. Agency for Healthcare Research and Quality. About the National Quality Strategy. Accessed 4 August 2025. Available from https://www.ahrq.gov/workingforquality/about/nqs-fact-sheets/nqs-fact-sheet-0214.html
80. Odugbesan O, Mungmode A, Rioles N, et al.; T1D Exchange Quality Improvement Collaborative. Increasing continuous glucose monitoring use for non-Hispanic Black and Hispanic people with type 1 diabetes: results from the T1D Exchange Quality Improvement Collaborative Equity Study. Clin Diabetes 2024;42:40–48
81. Odugbesan O, Wright T, Jones N-HY, et al.; T1D Exchange Quality Improvement Collaborative. Increasing social determinants of health screening rates among six endocrinology centers across the United States: results from the T1D Exchange Quality Improvement Collaborative. Clin Diabetes 2024;42:49–55
82. Prahalad P, Ebekozien O, Alonso GT, et al.; T1D Exchange Quality Improvement Collaborative Study Group. Multi-clinic quality improvement initiative increases continuous glucose monitoring use among adolescents and young adults with type 1 diabetes. Clin Diabetes 2021;39:264–271
83. National Institute of Diabetes and Digestive and Kidney Diseases. Diabetes for Health Professionals. Accessed 8 August 2025. Available from https://www.niddk.nih.gov/health-information/professionals/clinical-tools-patient-management/diabetes
84. O'Connor PJ, Sperl-Hillen JM, Fazio CJ, Averbeck BM, Rank BH, Margolis KL. Outpatient diabetes clinical decision support: current status and future directions. Diabet Med 2016;33:734–741
85. Forsythe LP, Carman KL, Szydlowski V, et al. Patient engagement in research: early findings from The Patient-Centered Outcomes Research Institute. Health Aff (Millwood) 2019;38:359–367
86. Marino M, Angier H, Springer R, et al. The Affordable Care Act: effects of insurance on diabetes biomarkers. Diabetes Care 2020;43:2074–2081
87. Centers for Disease Control and Prevention. Health Insurance Coverage: Early Release of Estimates From the National Health Interview Survey, January–June 2024. Accessed 4 August 2025. Available from https://www.cdc.gov/nchs/data/nhis/earlyrelease/insur202412.pdf
88. Doucette ED, Salas J, Scherrer JF. Insurance coverage and diabetes quality indicators among patients in NHANES. Am J Manag Care 2016; 22:484–490
89. Kaiser Family Foundation. 2021 Employer Health Benefits Survey. Accessed 8 August 2025. Available from https://www.kff.org/report-section/ehbs-2021-summary-of-findings/#:~:text=Twenty-eight%20percent%20of%20covered%20workers%20are%20enrolled%20in,as%20an%20indemnity%29%20plan%20%5B%20Figure%20D%20%5D
90. U.S. Bureau of Labor Statistics. Employee benefits—high deductible health plans and health savings accounts. Accessed 8 August 2025. Available from https://www.bls.gov/ebs/factsheets/high-deductible-health-plans-and-health-savings-accounts.htm
91. Garabedian LF, Zhang F, LeCates R, Wallace J, Ross-Degnan D, Wharam JF. Trends in high deductible health plan enrolment and spending among commercially insured members with and without chronic conditions: a Natural Experiment for Translation in Diabetes (NEXT-D2) study. BMJ Open 2021;11:e044198
92. Wu YM, Huang J, Reed ME. Association between high-deductible health plans and engagement in routine medical care for type 2 diabetes in a privately insured population: a propensity score-matched study. Diabetes Care 2022;45:1193–1200
93. Jiang DH, Herrin J, Van Houten HK, McCoy RG. Evaluation of high-deductible health plans and acute glycemic complications among adults with diabetes. JAMA Netw Open 2023;6:e2250602
94. Lu CY, Argetsinger S, Lakoma M, Zhang F, Wharam JF, Ross-Degnan D. Effects of adopting preventive drug lists on medication costs and disparities by income over 2 years: a Natural Experiment for Translation in Diabetes (NEXT-D) study. Diabetes Care 2025;48:341–352
95. McCoy RG, Swarna KS, Jiang DH, et al. Enrollment in high-deductible health plans and incident diabetes complications. JAMA Netw Open 2024;7:e243394
96. Gibson DM. Estimates of the percentage of US adults with diabetes who could be screened for diabetic retinopathy in primary care settings. JAMA Ophthalmol 2019;137:440–444
97. Gaffney A, Himmelstein DU, Woolhandler S. Prevalence and correlates of patient rationing of insulin in the United States: a national survey. Ann Intern Med 2022;175:1623–1626
98. Cefalu WT, Dawes DE, Gavlak G, et al.; Insulin Access and Affordability Working Group. Insulin Access and Affordability Working Group: conclusions and recommendations. Diabetes Care 2018;41:1299–1311
99. Centers for Medicare & Medicaid Services. Part D Senior Savings Model. Accessed 8 August 2025. Available from https://www.cms.gov/priorities/innovation/innovation-models/part-d-savings-model
100. Baig K, Dusetzina SB. Premiums and out-of-pocket spending for long-acting insulins under the Medicare Part D Senior Savings Model. JAMA 2022;328:2161–2162
101. Baig KA, Fry CE, Buntin MB, Powers AC, Dusetzina SB. Insulin out-of-pocket spending caps and employer-sponsored insurance: changes in out-of-pocket and total costs for insulin and healthcare. Health Serv Res 2025:e14656
102. American Diabetes Association. State Insulin Copay Caps. Accessed 8 August 2025. Available from https://diabetes.org/tools-resources/affordable-insulin/state-insulin-copay-caps
103. Suran M. All 3 major insulin manufacturers are cutting their prices—here's what the news means for patients with diabetes. JAMA 2023; 329:1337–1339
104. American Diabetes Association. Insulin Cost and Affordability. Leading the Fight for Insulin Affordability. Accessed 8 August 2025. Available from https://diabetes.org/tools-support/insulin-affordability
105. McCoy RG, Van Houten HK, Karaca-Mandic P, Ross JS, Montori VM, Shah ND. Second-line

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therapy for type 2 diabetes management: the treatment/benefit paradox of cardiovascular and kidney comorbidities. Diabetes Care 2021;44: 2302–2311
106. McCoy RG, Van Houten HK, Deng Y, et al. Comparison of diabetes medications used by adults with commercial insurance vs Medicare Advantage, 2016 to 2019. JAMA Netw Open 2021; 4:e2035792
107. Benning TJ, Heien HC, Herges JR, Creo AL, Al Nofal A, McCoy RG. Glucagon fill rates and cost among children and adolescents with type 1 diabetes in the United States, 2011-2021. Diabetes Res Clin Pract 2023;206:111026
108. Herges JR, Haag JD, Kosloski-Tarpenning KA, Mara KC, McCoy RG. Gaps in glucagon fills among commercially insured patients receiving a glucagon prescription. Diabetes Res Clin Pract 2023;200:110720
109. Patel PM, Thomas D, Liu Z, Aldrich-Renner S, Clemons M, Patel BV. Systematic review of disparities in continuous glucose monitoring and insulin pump utilization in the United States: key themes and evidentiary gaps. Diabetes Obes Metab 2024;26:4293–4301
110. Eberly LA, Yang L, Essien UR, et al. Racial, ethnic, and socioeconomic inequities in glucagon-like peptide-1 receptor agonist use among patients with diabetes in the US. JAMA Health Forum 2021;2:e214182
111. Essien UR, Singh B, Swabe G, et al. Association of prescription co-payment with adherence to glucagon-like peptide-1 receptor agonist and sodium-glucose cotransporter-2 inhibitor therapies in patients with heart failure and diabetes. JAMA Netw Open 2023;6: e2316290
112. Taha MB, Valero-Elizondo J, Yahya T, et al. Cost-related medication nonadherence in adults with diabetes in the United States: the National Health Interview Survey 2013–2018. Diabetes Care 2022;45:594–603
113. Centers for Medicare & Medicaid Services. CMS Framework for Health Equity 2022–2032. Accessed 8 August 2025. Available from https://www.cms.gov/files/document/cms-framework-health-equity.pdf
114. Washington AE, Lipstein SH. The Patient-Centered Outcomes Research Institute–promoting better information, decisions, and health. N Engl J Med 2011;365:e31
115. Dixon B, Peña M-M, Taveras EM. Lifecourse approach to racial/ethnic disparities in childhood obesity. Adv Nutr 2012;3:73–82
116. Egede LE, Walker RJ, Linde S, et al. Nonmedical interventions for type 2 diabetes: evidence, actionable strategies, and policy opportunities. Health Aff (Millwood) 2022;41: 963–970
117. Jiang DH, O'Connor PJ, Huguet N, Golden SH, McCoy RG. Modernizing diabetes care quality measures. Health Aff (Millwood) 2022;41:955–962
118. Joint Commission. R3 Report. Requirement, Rationale, Reference. Accessed 11 September 2025. Available from https://digitalassets.jointcommission.org/api/public/content/e57b92c0e6854a03b27302663b5c1084?v=cec3044d&_gl=1*1xqrmfh*_ga*Lg.*_ga_K31T0BHP4T*czE3NTQ0OTIwNDYkbzEkZzEkdDE3NTQ0OTIxNTUkajU5JGwwJGgw
119. National Committee for Quality Assurance. Healthcare Effectiveness Data and Information Set (HEDIS-MY) 2024. Accessed 6 August 2025. Available from https://www.ncqa.org/wp-content/uploads/HEDIS-MY-2024-Measure-Description.pdf
120. Centers for Medicare & Medicaid Services. Quality Payment Program. Explore Measures & Activities. Accessed 6 August 2025. Available from https://qpp.cms.gov/mips/explore-measures
121. Ebekozien O, Mungmode A, Buckingham D, et al. Achieving equity in diabetes research: borrowing from the field of quality improvement using a practical framework and improvement tools. Diabetes Spectr 2022;35:304–312
122. Odugbesan O, Addala A, Nelson G, et al. Implicit racial-ethnic and insurance-mediated bias to recommending diabetes technology: insights from T1D Exchange Multicenter Pediatric and Adult Diabetes Provider Cohort. Diabetes Technol Ther 2022;24:619–627
123. Centers for Medicare & Medicaid Services. Ensuring Proper Use of Electronic Health Record Features and Capabilities. 2016. Accessed 6 August 2025. Available from https://www.cms.gov/files/document/ehrdecisiontable062816pdf
124. U.S. Department of Health and Human Services. Secretary's Advisory Committee on National Health Promotion and Disease Prevention Objectives for 2030. Accessed 6 August 2025. Available from https://health.gov/healthypeople
125. National Academy of Sciences. A Framework for Educating Health Professionals to Address the Social Determinants of Health. 2016. Accessed 6 August 2025. Available from https://www.ncbi.nlm.nih.gov/pubmed/27854400
126. Zakaria NI, Tehranifar P, Laferrère B, Albrecht SS. Racial and ethnic disparities in glycemic control among insured US adults. JAMA Netw Open 2023;6:e2336307
127. Hershey JA, Morone J, Lipman TH, Hawkes CP. Social determinants of health, goals and outcomes in high-risk children with type 1 diabetes. Can J Diabetes 2021;45:444–450.e1
128. Walker RJ, Strom Williams J, Egede LE. Influence of race, ethnicity and social determinants of health on diabetes outcomes. Am J Med Sci 2016;351:366–373
129. Hashemi R, Rabizadeh S, Yadegar A, et al. High prevalence of comorbidities in older adult patients with type 2 diabetes: a cross-sectional survey. BMC Geriatr 2024;24:873
130. Huang ES. Management of diabetes mellitus in older people with comorbidities. BMJ 2016; 353:i2200
131. O'Gurek DT, Henke C. A practical approach to screening for social determinants of health. Fam Pract Manag 2018;25:7–12
132. Yan AF, Chen Z, Wang Y, et al. Effectiveness of social needs screening and interventions in clinical settings on utilization, cost, and clinical outcomes: a systematic review. Health Equity 2022;6:454–475
133. Rabbitt MP, Hales LJ, Burke MP, Coleman-Jensen A. *Household Food Security in the United States in 2022 (Report No. ERR-325).* U.S. Department of Agriculture, Economic Research Service, 2023. Accessed 11 September 2025. Available from https://search.nal.usda.gov/discovery/fulldisplay?context=L&vid=01NAL_INST:MAIN&docid=alma-9916411232407426
134. Kirby JB, Bernard D, Liang L. The prevalence of food insecurity is highest among Americans for whom diet is most critical to health. Diabetes Care 2021;44:e131–e132
135. Levi R, Bleich SN, Seligman HK. Food insecurity and diabetes: overview of intersections and potential dual solutions. Diabetes Care 2023; 46:1599–1608
136. Alawode O, Humble S, Herrick CJ. Food insecurity, SNAP participation and glycemic control in low-income adults with predominantly type 2 diabetes: a cross-sectional analysis using NHANES 2007-2018 data. BMJ Open Diabetes Res Care 2023;11:e003205
137. Hager ER, Quigg AM, Black MM, et al. Development and validity of a 2-item screen to identify families at risk for food insecurity. Pediatrics 2010;126:e26–e32
138. Little M, Rosa E, Heasley C, Asif A, Dodd W, Richter A. Promoting healthy food access and nutrition in primary care: a systematic scoping review of food prescription programs. Am J Health Promot 2022;36:518–536
139. Berkowitz SA, Meigs JB, DeWalt D, et al. Material need insecurities, control of diabetes mellitus, and use of health care resources: results of the Measuring Economic Insecurity in Diabetes study. JAMA Intern Med 2015;175:257–265
140. Reid LA, Mendoza JA, Merchant AT, et al. Household food insecurity is associated with diabetic ketoacidosis but not severe hypoglycemia or glycemic control in youth and young adults with youth-onset type 2 diabetes. Pediatr Diabetes 2022;23:982–990
141. White BM, Logan A, Magwood GS. Access to diabetes care for populations experiencing homelessness: an integrated review. Curr Diab Rep 2016;16:112
142. Stahre M, VanEenwyk J, Siegel P, Njai R. Housing insecurity and the association with health outcomes and unhealthy behaviors, Washington State, 2011. Prev Chronic Dis 2015;12:E109
143. Bernstein RS, Meurer LN, Plumb EJ, Jackson JL. Diabetes and hypertension prevalence in homeless adults in the United States: a systematic review and meta-analysis. Am J Public Health 2015;105:e46–e60
144. Montgomery AE, Fargo JD, Kane V, Culhane DP. Development and validation of an instrument to assess imminent risk of homelessness among veterans. Public Health Rep 2014;129:428–436
145. Baxter AJ, Tweed EJ, Katikireddi SV, Thomson H. Effects of housing first approaches on health and well-being of adults who are homeless or at risk of homelessness: systematic review and meta-analysis of randomised controlled trials. J Epidemiol Community Health 2019;73:379–387
146. Al-Rousan T, AlHeresh R, Saadi A, et al. Epidemiology of cardiovascular disease and its risk factors among refugees and asylum seekers: systematic review and meta-analysis. Int J Cardiol Cardiovasc Risk Prev 2022;12:200126
147. Jaung MS, Willis R, Sharma P, et al. Models of care for patients with hypertension and diabetes in humanitarian crises: a systematic review. Health Policy Plan 2021;36:509–532
148. Olson RM, Nolan CP, Limaye N, Osei M, Palazuelos D. National prevalence of diabetes and barriers to care among U.S. farmworkers and association with migrant worker status. Diabetes Care 2023;46:2188–2192
149. U.S. Department of Agriculture, National Agricultural Statistics Service. Census of Agriculture. 2022 Census Full Report. Accessed 6 August 2025. Available from https://www.nass.usda.gov/Publications/AgCensus/2022/index.php#full_report

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150. Soto S, Yoder AM, Aceves B, Nuño T, Sepulveda R, Rosales CB. Determining regional differences in barriers to accessing health care among farmworkers using the National Agricultural Workers Survey. J Immigr Minor Health 2023;25: 324–330
151. Thonon F, Perrot S, Yergolkar AV, et al. Electronic tools to bridge the language gap in health care for people who have migrated: systematic review. J Med Internet Res 2021;23: e25131
152. U.S. Department of Health & Human Services. National Standards for Culturally and Linguistically Appropriate Services (CLAS) in Health and Health Care. Accessed 6 August 2025. Available from https://thinkculturalhealth.hhs.gov/assets/pdfs/EnhancedNationalCLASStandards_Revised_June2025.pdf
153. Centers for Disease Control and Prevention. Health Literacy. Talking Points About Health Literacy. Accessed 06 August 2025. Available from https://www.cdc.gov/health-literacy/php/about/tell-others.html
154. Nutbeam D, Lloyd JE. Understanding and responding to health literacy as a social determinant of health. Annu Rev Public Health 2021;42:159–173
155. Marciano L, Camerini A-L, Schulz PJ. The role of health literacy in diabetes knowledge, self-care, and glycemic control: a meta-analysis. J Gen Intern Med 2019;34:1007–1017
156. Schaffler J, Leung K, Tremblay S, et al. The effectiveness of self-management interventions for individuals with low health literacy and/or low income: a descriptive systematic review. J Gen Intern Med 2018;33:510–523
157. Butayeva J, Ratan ZA, Downie S, Hosseinzadeh H. The impact of health literacy interventions on glycemic control and self-management outcomes among type 2 diabetes mellitus: a systematic review. J Diabetes 2023;15:724–735
158. Baumeister A, Aldin A, Chakraverty D, et al. Interventions for improving health literacy in migrants. Cochrane Database Syst Rev 2023;11: CD013303
159. Schapira MM, Fletcher KE, Gilligan MA, et al. A framework for health numeracy: how patients use quantitative skills in health care. J Health Commun 2008;13:501–517
160. Agency for Healthcare Research and Quality. Clinical-community linkages. Accessed 6 August 2025. Available from https://archive.ahrq.gov/ncepcr/tools/community/index.html
161. Herges JR, Matulis JC, Kessler ME, Ruehmann LL, Mara KC, McCoy RG. Evaluation of an enhanced primary care team model to improve diabetes care. Ann Fam Med 2022;20:505–511
162. Kasper AL, Myers LA, Carlson PN, et al. Diabetes management for community paramedics: development and implementation of a novel curriculum. Diabetes Spectr 2022;35:367–376
163. Evans J, White P, Ha H. Evaluating the effectiveness of community health worker interventions on glycaemic control in type 2 diabetes: a systematic review and meta-analysis. Lancet 2023;402(Suppl 1):S40
164. Ducharme-Smith A, Juntunen M, Espinoza A, et al. Pilot pragmatic trial of a community paramedic diabetes self-management education program for adults with diabetes. Diabetes Care 2025;48:2089–2094
165. Azmiardi A, Murti B, Febrinasari RP, Tamtomo DG. The effect of peer support in diabetes self-management education on glycemic control in patients with type 2 diabetes: a systematic review and meta-analysis. Epidemiol Health 2021;43:e2021090
166. Fisher EB, Boothroyd RI, Elstad EA, et al. Peer support of complex health behaviors in prevention and disease management with special reference to diabetes: systematic reviews. Clin Diabetes Endocrinol 2017;3:4
167. Foster G, Taylor SJC, Eldridge SE, Ramsay J, Griffiths CJ. Self-management education programmes by lay leaders for people with chronic conditions. Cochrane Database Syst Rev 2007: CD005108
168. Piatt GA, Rodgers EA, Xue L, Zgibor JC. Integration and utilization of peer leaders for diabetes self-management support: results from Project SEED (Support, Education, and Evaluation in Diabetes). Diabetes Educ 2018;44:373–382
169. Rosenthal EL, Rush CH, Allen CG. Understanding scope and competencies: a contemporary look at the United States community health worker field: progress report of the Community Health Worker (CHW) Core Consensus (C3) Project: building national consensus on CHW core roles, skills, and qualities. CHW Central, 2016. Accessed 6 August 2025. Available from https://files.ctctcdn.com/a907c850501/1c1289f0-88cc-49c3-a238-66def942c147.pdf
170. Guide to Community Preventive Services. Community health workers help patients manage diabetes. Updated 25 October 2022. Accessed 06 August 2025. Available from https://www.thecommunityguide.org/content/community-health-workers-help-patients-manage-diabetes
171. Cuellar AE, Calonge BN. The Community Preventive Services Task Force: 25 years of effectiveness, economics, and equity. Am J Prev Med 2022;62:e371–e373

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